

Introduction

This document specifies all parts that teams are permitted to use when constructing their robots. The AIRCER staff will be responsible for managing and distributing the listed parts.

Please note that certain parts are only available in limited quantities. To prevent shortages, it is recommended that teams manage their resources responsible. Broken or defective parts can be exchanged for a replacement. Teams are allowed to acquire additional or replacement parts at their own expense.

Table of Contents

Introduction	1
AIRCER Junior Conditions	2
Metal Parts	2
KIPR Metal	2
Screws, Nuts and Standoffs	4
Electronics	7
Sensors	7
Motion	8
Controller	8
MISC.	9
Robot Design	9
AIRCER Senior:.....	9
AIRCER Junior:	10
LEGO ®	10
3D Print Rules	10
Additional Materials	11
Kit Rules	12
Second Bot Rule	14

AIRCER Junior Conditions

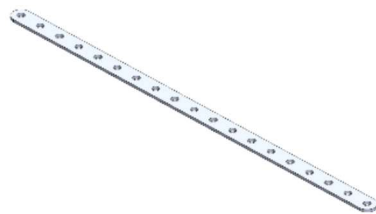
Teams participating in the AIRCER Junior Division are not subject to the part limitations specified in this document and are encouraged to use additional materials when preparing for the competition. However, these teams are **NOT** exempt from the Kit Rules (excluding rules 1 and 2), and any additional material must adhere to these rules.

Metal Parts

KIPR Metal

* ... Parts may be bend

Strap 1 x 19 *



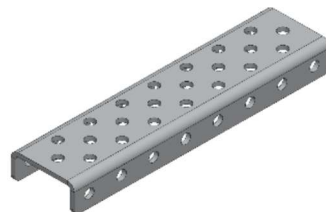
2

Strap 1 x 10 *



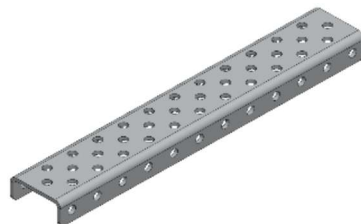
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Channel 3 x 8



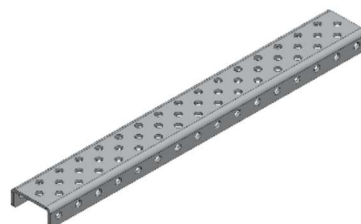
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Channel 3 x 12

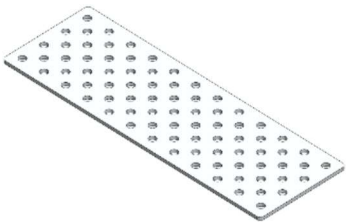
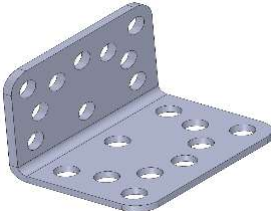
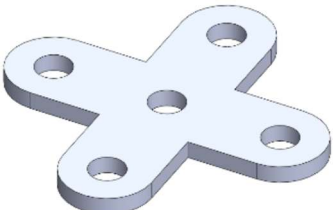
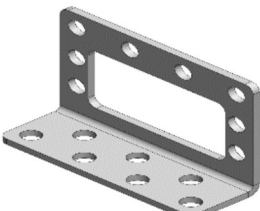
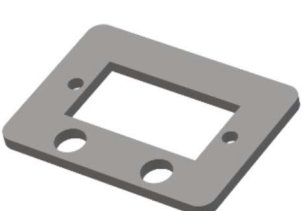


1

Channel 3 x 16



1

Plate 4 x 12 *		2
D Bracket		1
Angle Bracket		1
Plus Plate *		1
Servo Bracket		4
Round Servo Horn		2
1 x 5 Servo Horn		2
Micro Servo Mount		2

Screws, Nuts and Standoffs

Screw 8-32 x 1/4"
(black)



20

Screw 8-32 x 1/2"
(black)



40

Screw 8-32 x 3/4"
(black)



40

Nut 8-32 Keps
(black)



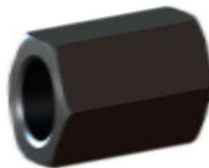
80

Nut 8-32 Nylock
(black)



20

Standoff 8-32 x 1/2"

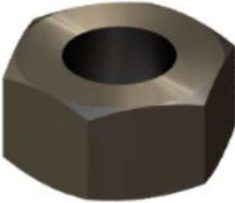


6

Standoff 8-32 x 1"



3

Standoff 8-32 x 2"		1
Standoff 8-32 x 3"		1
Short Silver Servo Horn Screw (M3)		4
18-8 Washer		4
Silver Servo Horn Nut (M3)		6
Silver Servo Horn Machine Screw (M3)		6
Threaded Rod 8-32 x 1"		4
Threaded Rod 8-32 x 2"		1

Threaded Rod 8-32
x 3"



1

Threaded Rod 8-32
x 6"



1

Brass Screws



10

2-56 1" Caster
Screw



2

2-56 3/8" Micro
Servo Mount Screw



5

M3 x 0.5" x 6"
Mecanum Wheel
Screw



4

Electronics

Sensors

Analog

ET
(Rangefinder)



1

Tophat
(Reflectance)



2

Small Tophat
(Reflectance)



1

Light



1

Linear Slide



1

Digital

Lever



1

Small Touch



2

Large Touch



2

Motion

Servo
(MG995)



or



4

Black Gear Motor



6

Micro Servo
(SG90)



or



2

Controller

Wombat



1

Black Camera



1

MISC.

Servo Extension
Cable



2

Sensor Extension
Cable



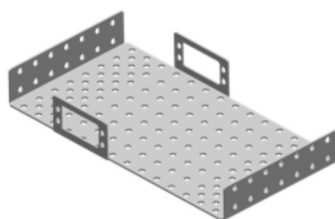
2

Robot Design

Teams must use the following parts as the base of their robot:

AIRCER Senior:

Chassis



1

Mecanum Wheel



4

AIRCER Junior:

Create 2

or

Create 3



1

LEGO®

Teams are permitted to use LEGO® bricks of any number and color, provided they are part of the LEGO® Technic subcategory.

Should your team require an overview of parts that may be useful in the construction of your robots, please refer to the current Botball LEGO® Parts List.

3D Print Rules

1. Only PLA or PETG may be used to print parts.
 - Parts must be printed using a grayscale-colored PLA or PETG material.
 - PLA or PETG material must only contain PLA or PETG.
 - PLA or PETG must be printed and not in its raw unprinted form.
2. The number of 3D printed parts may not exceed 4 total between all robots at the table.
 - A part is a single static piece.
 - If parts are connected using material in an attempt to circumvent this rule, they will not be permitted.
 - If there are moving parts in a 3D printed assembly, each movable part counts towards the 4 total parts. Examples include, a chain with 10 links in one print would still count as 10 parts, two or more parts tethered by a piece of material will be counted as two or more parts.
 - Only 4 3D printed parts for use on the robot may be brought to the On-Deck area, excluding the required duplicate identical copies for measurement and any jigs being used for positioning the robots.

3. A single part cannot exceed the print volume of 220 mm x 220 mm x 60 mm.
 - If using a part on a robot, a second identical copy of the part may be required for
 - judges to check the measurements. If a second identical copy of the part cannot be provided to the judge, the robot it is attached to may be disqualified for the round.
 - A second identical copy of the part may be needed for onsite presentations.
 - A box for measuring parts will be provided throughout both practice and tournament for teams to check part dimensions.
 - At tournaments, judges may have a box of the correct dimensions that a part must fit in.
4. A STL file must be submitted for each part that may be in use on a robot.
 - At the tournament the teams must use the 3D-Prints approved by the 3D-Printing-Team. (Parts may only be printed in the AIRlab)

Additional Materials

Teams may add the following materials to their robot:

- String/thread/fishing line: Up to 100 cm, max 1 mm diameter, non-metallic, not for offensive use (e.g., entanglement).
- Paper: One A4 (210×297 mm) sheet.
- Foam board/corrugated plastic: One sheet, max 5mm thick, max A4 footprint.
- Rubber bands: Up to 10 total, standard size #19 (3.5" × 1/16" × 1/32").
- Paper clips: Up to 10 metal clips (1"–1.5" in length); may be bent but not cut or electrically used.
- Coins: Up to 250 g (approx. 108 1-Cent-Coins) used solely as counterweights. Coins may be rolled for convenience.
- Zip ties: Up to 10 white or black zip ties may be used for any purpose.

Kit Rules

1. **Sensor Usage:**

- Sensors from the 2017–2026 Botball kits may be used, as long as they do not exceed the type or number included in the 2026 AIRCER kit.

2. **KIPR Metal Plate (KMP):**

- Only the steel chassis from the KMP may be used in robot construction.

3. **Mounting Adhesives:**

- Small removable mounting dots/strips (e.g., Glue Dots, UGlu, Scotch Brand Restickable, or blue tack) may be used for construction purposes only. These must not:
 - Be exposed to stick to the game board or game elements
 - Be used to contact the opponent's entry
- No other adhesives (e.g., hot melt glue) are allowed.

4. **Wire Management:**

- Tape: (non-game piece colors only) may be used solely for managing or labeling wires—not for construction.
- Twist ties: may also be used for managing wires only.

5. **Excess Adhesive Use:**

- Judges may require removal of any excessive adhesive. Mechanical construction methods are preferred and should be prioritized.

6. **Servo Accessories:**

- Supplied accessories (e.g., grommets, screws, washers) may only be used to mount components to the servo horn.

7. Mounting Motors and Servos:

- These may be attached to structural pieces using the supplied machine screws.

8. Sensor Modifications:

- Teams may trim connector potting material to assist with mounting or insertion. Damaged components must be replaced at the team's expense.

9. Servo Horn Modifications:

- Plastic servo horns may be trimmed. Damaged horns must be replaced at the team's expense.

10. Paper & Foam Usage Verification:

- If a team's use of paper, foam board, or corrugated plastic appears excessive or unclear, the head judge may require a single-template layout to prove all material originating from one sheet of each type.
- Only black or white material is allowed.
- Only grayscale printing permitted (e.g., team sponsor logos, QR codes).
- These materials may only be mounted using legal kit parts or allowable adhesives.

11. Rubber Band Rules:

- May not be glued or melted
- May be cut, but the total number (whole or cut) is limited to 10
- Any combination of whole and cut bands must not exceed 10 total

12. Light Guide Materials: Light guides (for sensors only) may be made from:

- Soda straws, paper, electrical tape, or foil

- May be shielded using tape but not structurally supported or used for manipulation. These materials are in addition to the allowable kit parts.

13. Sensor Shielding Restrictions:

- External shielding of sensors (e.g., by standing between the robot and the audience) is not allowed. Teams must calibrate and orient their sensors properly. Teams using cameras may request individuals wearing clothing matching game object colors to stand back from the table.

14. Lego Part Modifications:

- LEGO® parts may not be physically modified.
- Exception: Screws may be used through axle holes.

15. Metal Part Limitations:

- Metal parts may not be cut or broken.
- Only listed straps and plates may be bent (brackets may not).
- Metal edges may be filed or sanded for safety.
- Metal surfaces may be sanded to reduce friction.

Second Bot Rule

Teams have the opportunity to earn the right to build and use an additional robot by completing a challenge set by the Judge Team. The requirements for completing the challenge are subject to change. Only a Head Judge has the authority to approve a successful challenge.

The materials permitted for the construction of an additional robot vary according to team division. Further information will be provided upon completion of the challenge.